Durban’s Pothole Challenge – Scale & Urgency

9,462 km

Durban’s road network

3.12 million m²

pothole backlog in KwaZulu-Natal

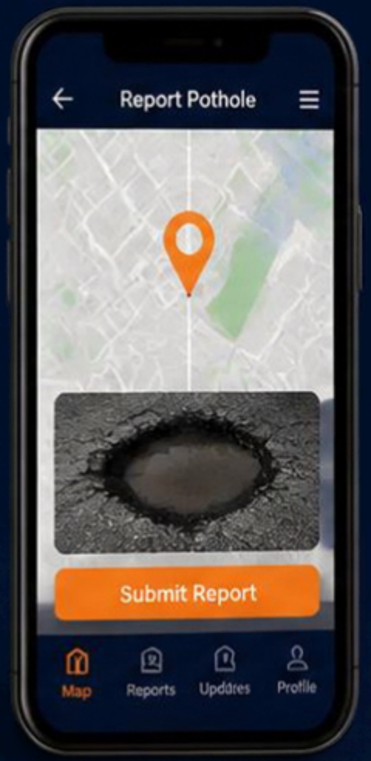
R216 million

allocated in 2026 for pothole repairs in KZN

Poor roads put safety, mobility and the economy at risk.



Our Smart Pothole Detection Solution



COMMUNITY APP

Easy pothole reporting with photo and GPS location

Real-time status updates and map visualization



SMART SENSOR HARDWARE

ESP32 + accelerometer + GPS + Bluetooth detects road impacts automatically

User confirms pothole before report submission



CENTRALISED DATABASE

Stores location, severity, verification, and repair tracking

How It Works

From Detection to Repair Tracking

01



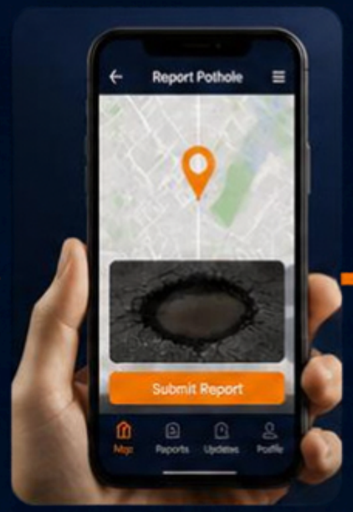
Sensor detects impact and logs location

02



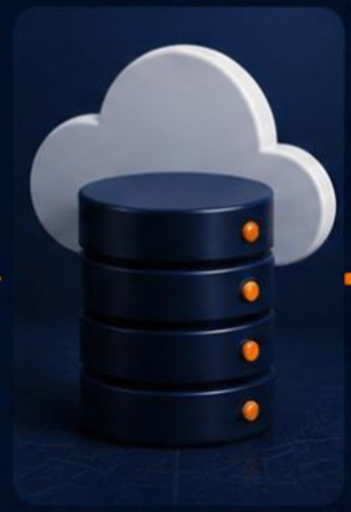
Data sent via Bluetooth to smartphone app

03



User reviews details and submits report

04



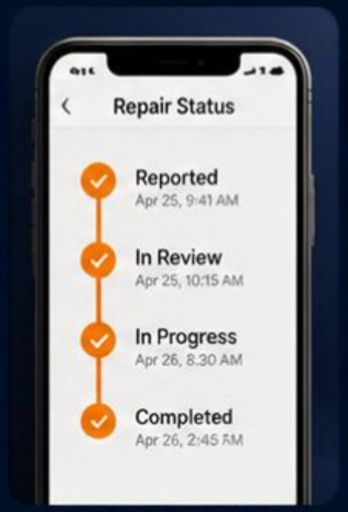
Report stored in central database

05



Authorities notified and repairs scheduled

06



Repair progress updated and tracked in real time

Community & Economic Benefits

01

STRONGER ROAD SAFETY

Better hazard detection enables quicker responses

02

GREATER MOBILITY & ACCESS

Improved roads connect people to work, school, healthcare, and transport

03

HEALTH & WELLBEING

Safer travel and easier access to essential services

04

ECONOMIC IMPACT

Boosts local SMMEs and creates jobs
N2/N3 projects employed 7,430 labourers and involved 576 SMMEs

05

COMMUNITY EMPOWERMENT

Residents actively participate in infrastructure monitoring



Funding & Partnership Opportunity



Join a technology-driven, community-centred road safety initiative

Strengthen and complement existing municipal and provincial road efforts

Scale impact for safer roads, better mobility, and a stronger local economy

Let’s build safer communities together